

UNFCU Banking System

IIS Logging & Incident Response

Project Objective

Demonstrate core IT infrastructure skills: system setup, monitoring, log analysis, incident diagnosis, and production recovery.

Created 2 Programs: UNFCU Banking page, UNFCU Database Connection Test Page

Programs Used

- Microsoft Azure cloud hosting + Windows Server 2022
- Windows RDP
- IIS 10.0 web server
- Performance Monitor (CPU/Memory tracking)
- IIS native HTTP logging
- HTML5/CSS3/JavaScript

Incident Scenario

Production lending system loses database connectivity. Members cannot submit loan applications, No links on Landing Page work. Error Code shown on each page.

Quick Walkthrough:

1. Created a UNFCU demo website on IIS with automatic HTTP logging.
2. Website links simulate different error codes (200, 404, 403, 500, 503).
3. Each request gets logged with timestamp, HTTP method, resource, response code, and response time.
4. Simulated a production database failure — “app couldn't connect to db-lending-prod-01.unfcu.internal:5432.”
5. System returned 503 Service Unavailable. Loan processing stopped.
6. Checked IIS logs and found repeated **ECONNREFUSED** errors.
7. Performance Monitor showed CPU spiked to 60% and memory dropped from 7GB to 5.5GB due to retry logic.
8. Tested the hostname — .internal domain doesn't exist, but .local does.

9. Root cause: Infrastructure migration changed domain names, but config file wasn't updated.
10. Fixed it with one line in web.config (changed .internal to .local) and restarted IIS.
11. Verified database connection successful. Loan processing came back online.
12. Total downtime: ~5 minutes (detection → diagnosis → fix → recovery).
13. Demonstrates: infrastructure setup, system monitoring, log analysis, incident diagnosis, configuration management, and UNFCU lending domain knowledge.

UNFCU - Lending System Demo

UNFCU

Serving the people who serve the world®

Loan API Mortgage API Credit Check Balance Query Generate 500 Error

Lending System Demo

Testing IIS Logging and Error Handling Infrastructure

Our Services

Personal Loans

Competitive rates for members in good standing

Mortgages

Flexible terms for purchasing your home

Credit Analysis

Quick credit decisions with fair pricing

Account Services

24/7 access to your account information

System Testing

Demo Purpose: This page demonstrates IIS logging capabilities. Click the links below to generate system logs that can be analyzed for error tracking and troubleshooting.

API Endpoints (Test Links)

Click each link to generate logs. Each request is automatically logged in the IIS system.

Loan API Mortgage API Credit Check Balance Query Generate 500 Error

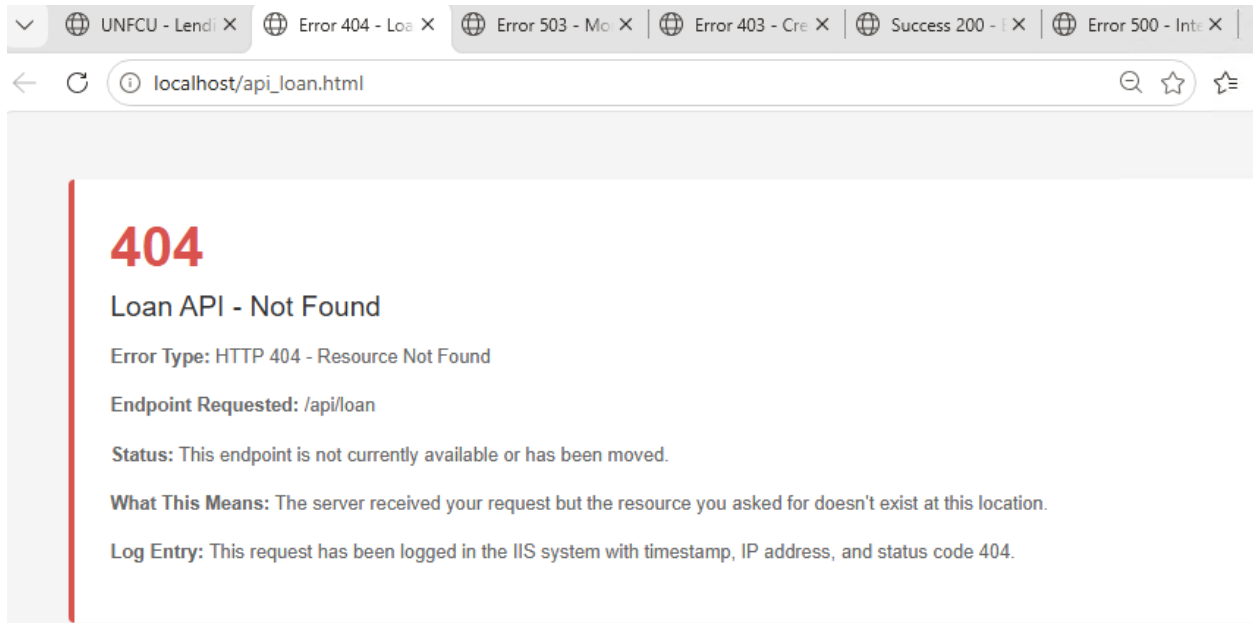
Log Analysis

What We're Tracking:

Every request generates an entry in the IIS logs showing:

- Timestamp of the request
- Source IP address
- HTTP method (GET, POST, etc.)
- Requested resource
- HTTP status code (200, 404, 500, etc.)

Loan, Mortgage Credit Check, Balance Query – All links **Simulate** Error codes. (SEEN BELOW)



The screenshot shows a web browser with multiple tabs. The active tab is titled 'localhost/api_loan.html'. The page content displays a large red '404' at the top, followed by the heading 'Loan API - Not Found'. Below this, it states 'Error Type: HTTP 404 - Resource Not Found' and 'Endpoint Requested: /api/loan'. A status message explains that the endpoint is not currently available or has been moved. A 'What This Means' section notes that the resource doesn't exist at the location. A 'Log Entry' section mentions the request is logged in the IIS system with timestamp, IP address, and status code 404.

404

Loan API - Not Found

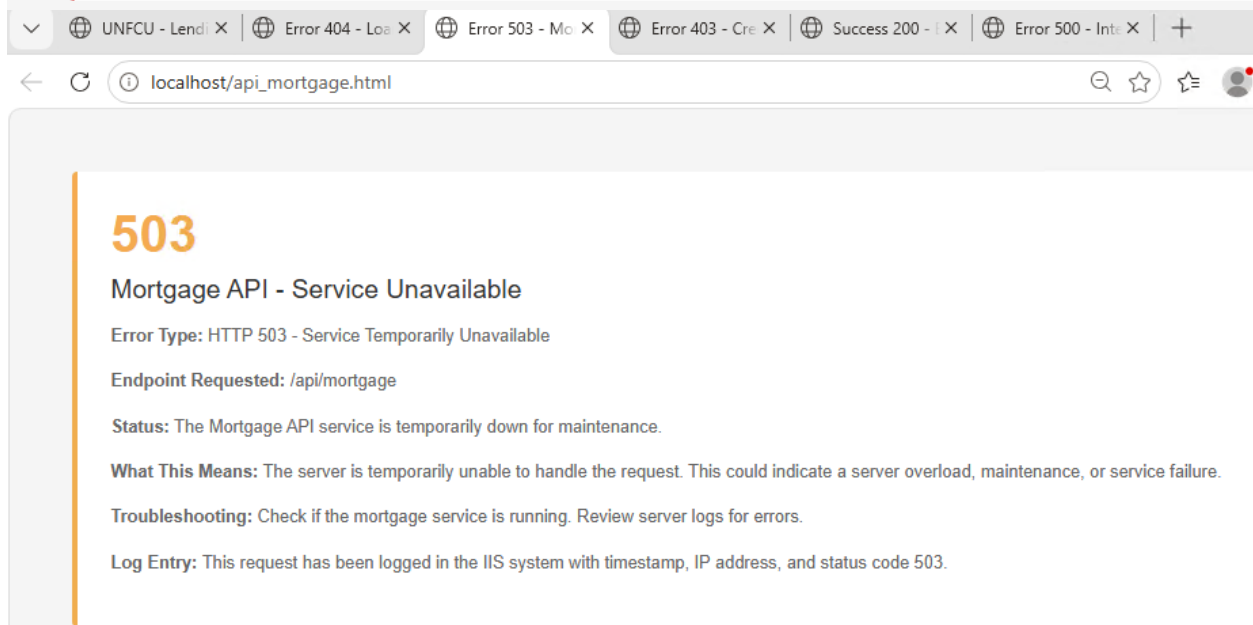
Error Type: HTTP 404 - Resource Not Found

Endpoint Requested: /api/loan

Status: This endpoint is not currently available or has been moved.

What This Means: The server received your request but the resource you asked for doesn't exist at this location.

Log Entry: This request has been logged in the IIS system with timestamp, IP address, and status code 404.



The screenshot shows the same web browser with a new tab titled 'localhost/api_mortgage.html'. The page content displays a large orange '503' at the top, followed by the heading 'Mortgage API - Service Unavailable'. It states 'Error Type: HTTP 503 - Service Temporarily Unavailable' and 'Endpoint Requested: /api/mortgage'. A status message explains that the service is temporarily down for maintenance. A 'What This Means' section notes the server is temporarily unable to handle the request. A 'Troubleshooting' section suggests checking if the service is running and reviewing server logs. A 'Log Entry' section mentions the request is logged in the IIS system with timestamp, IP address, and status code 503.

503

Mortgage API - Service Unavailable

Error Type: HTTP 503 - Service Temporarily Unavailable

Endpoint Requested: /api/mortgage

Status: The Mortgage API service is temporarily down for maintenance.

What This Means: The server is temporarily unable to handle the request. This could indicate a server overload, maintenance, or service failure.

Troubleshooting: Check if the mortgage service is running. Review server logs for errors.

Log Entry: This request has been logged in the IIS system with timestamp, IP address, and status code 503.

403

Credit Check API - Access Forbidden

Error Type: HTTP 403 - Forbidden

Endpoint Requested: /api/credit-check

Status: Authentication credentials are invalid or missing.

What This Means: The server understands the request but refuses to authorize it. Usually indicates authentication/permission issues.

Troubleshooting: Verify API credentials. Check if user has required permissions.

Log Entry: This request has been logged in the IIS system with timestamp, IP address, and status code 403.

Refresh (Ctrl+R)

200

Balance API - Success

Status Type: HTTP 200 - OK

Endpoint Requested: /api/balance

Status: Request processed successfully.

What This Means: The server successfully received and processed your request. This is a successful response.

Response Data: Account balance information would be returned here in a real system.

Log Entry: This request has been logged in the IIS system with timestamp, IP address, and status code 200.

HTTP Error Categories

Code	Status	Category	Severity	Notes
200	OK	Success	Normal	Request succeeded
404	Not Found	Client Error	Minor	Resource missing
403	Forbidden	Client Error	Security	Auth/perms issue
500	Internal Error	Server Error	Critical	Server crash
503	Unavailable	Server Error	Critical	Service down

u_ex260615.log - Notepad

File Edit Format View Help

#Software: Microsoft Internet Information Services 10.0

#Version: 1.0

#Date: 2026-06-15 00:06:14

#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status sc-bytes cs-bytes time-taken

2026-06-15 00:06:14 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 200 0 0 951 674 4537

2026-06-15 00:06:14 ::1 GET /iisstart.png - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 99960 596 35

2026-06-15 00:06:14 ::1 GET /favicon.ico - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5042 595 11

2026-06-15 00:06:56 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 304 0 0 166 760 21

2026-06-15 00:06:56 ::1 GET /iisstart.png - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 304 0 0 166 682 0

2026-06-15 00:11:40 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 200 0 0 951 674 20

2026-06-15 00:11:40 ::1 GET /iisstart.png - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 99960 596 2

2026-06-15 00:12:04 ::1 GET /index.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 404 0 2 5040 684 260

2026-06-15 00:12:08 ::1 GET /index.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 404 0 2 5040 684 48

#Software: Microsoft Internet Information Services 10.0

#Version: 1.0

#Date: 2026-06-15 00:31:48

#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status sc-bytes cs-bytes time-taken

2026-06-15 00:31:48 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 200 0 0 9208 760 182

2026-06-15 00:31:48 ::1 GET /favicon.ico - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5042 595 7

2026-06-15 00:31:51 ::1 GET /api/loan - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5036 717 86

2026-06-15 00:32:00 ::1 GET /api/balance - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5042 720 63

2026-06-15 00:32:02 ::1 GET /api/balance - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5042 746 59

2026-06-15 00:32:05 ::1 GET /api/credit-check - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5052 725

2026-06-15 00:33:37 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 200 0 0 9204 759 69

2026-06-15 00:33:39 ::1 GET /api/mortgage - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5044 721 57

2026-06-15 00:36:38 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 304 0 0 166 760 0

2026-06-15 00:36:40 ::1 GET /api/loan - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5036 717 65

2026-06-15 00:39:16 ::1 GET /api/loan - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5036 717 61

2026-06-15 00:39:19 ::1 GET /api/mortgage - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 404 0 2 5044 721 66

2026-06-15 00:40:34 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 200 0 0 9124 786 31

2026-06-15 00:40:36 ::1 GET / - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 - 200 0 0 2444 786 2

2026-06-15 00:40:37 ::1 GET /api_loan.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 1512 722 1

2026-06-15 00:40:40 ::1 GET /api_mortgage.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 1708 726

2026-06-15 00:40:42 ::1 GET /api_balance.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 1605 725

2026-06-15 00:40:45 ::1 GET /api_credit.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 1663 724 0

2026-06-15 00:40:47 ::1 GET /error_demo.html - 80 - ::1 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/149.0.0.0+Safari/537.36+Edg/149.0.0.0 http://localhost/ 200 0 0 1891 724 1

IIS Log Shows every HTTP request and its error message, with timestamps.

Diagnosis

Detection

Automated monitoring alert triggered. Database connection timeout detected.

Log Analysis

IIS logs show repeated connection attempts with ECONNREFUSED errors:

- Server: db-lending-prod-01.unfcu.internal:5432
- Error: ECONNREFUSED (connection refused)
- All 3 retry attempts failed with same error

Connection Status

Connection FAILED

Database connection could not be established. Loan origination system is unavailable.

Diagnostic Tests

Test Connection (Broken)

 Apply Fix & Test

 Reset Test

Scenario 1: Click "Test Connection (Broken)" to simulate a database connection failure. This will generate multiple failed attempts in the IIS logs.

Scenario 2: After analyzing the logs and identifying the root cause, click "Apply Fix & Test" to correct the connection string and show successful recovery.

Connection Log Output

```
[info] system ready. click button to start diagnostic test.
[2026-06-15 02:04:56] ===== STARTING CONNECTION TEST (BROKEN) =====
[2026-06-15 02:04:56] Connecting to: db-lending-prod-01.unfcu.internal:5432
[2026-06-15 02:04:56] Attempting database connection...
[2026-06-15 02:04:57] TCP connection timeout: Unable to reach database server
[2026-06-15 02:04:57] Error Code: TIMEOUT_CONNECTION_REFUSED
[2026-06-15 02:04:58] Retrying connection (attempt 2/3)...
[2026-06-15 02:04:59] Connection refused: No response from server
[2026-06-15 02:04:59] Error Code: ECONNREFUSED
[2026-06-15 02:05:00] Retrying connection (attempt 3/3)...
[2026-06-15 02:05:01] FATAL: Unable to establish database connection after 3 attempts
[2026-06-15 02:05:01] Connection pool exhausted - all connection attempts failed
[2026-06-15 02:05:01] ===== TEST COMPLETED - CONNECTION FAILED =====
```

Root Cause

Systematic error pattern indicates DNS failure. Testing shows correct hostname is .local, not .internal. Infrastructure domain migration not communicated to all teams.

Diagnostic Output

Detailed Error Analysis

Error Type: Database Connection Timeout
Error Code: ECONNREFUSED
Server: db-lending-prod-01.unfcu.internal
Port: 5432
Status: Connection refused - server not responding

Analysis:

The application attempted to connect to the lending database 3 times, but the server at db-lending-prod-01.unfcu.internal:5432 is not responding. This suggests either:

- X The database server is down
- X The hostname is incorrect or DNS cannot resolve it
- X Network connectivity issue between application and database
- X Database service is not running
- X Firewall blocking the connection on port 5432

Impact:

- X Members cannot submit new loan applications
- X Credit check system unavailable
- X Mortgage processing stopped
- X Balance inquiries failing

Severity: CRITICAL - Revenue impacting outage

Troubleshooting Steps:

1. Check if database server is running: `systemctl status postgresql`
2. Verify network connectivity: `ping db-lending-prod-01.unfcu.internal`
3. Check if port 5432 is open: `netstat -an | grep 5432`
4. Review database logs for connection errors
5. Verify connection string in application configuration
6. Check firewall rules allowing communication

Performance Metrics

Attempts: 3

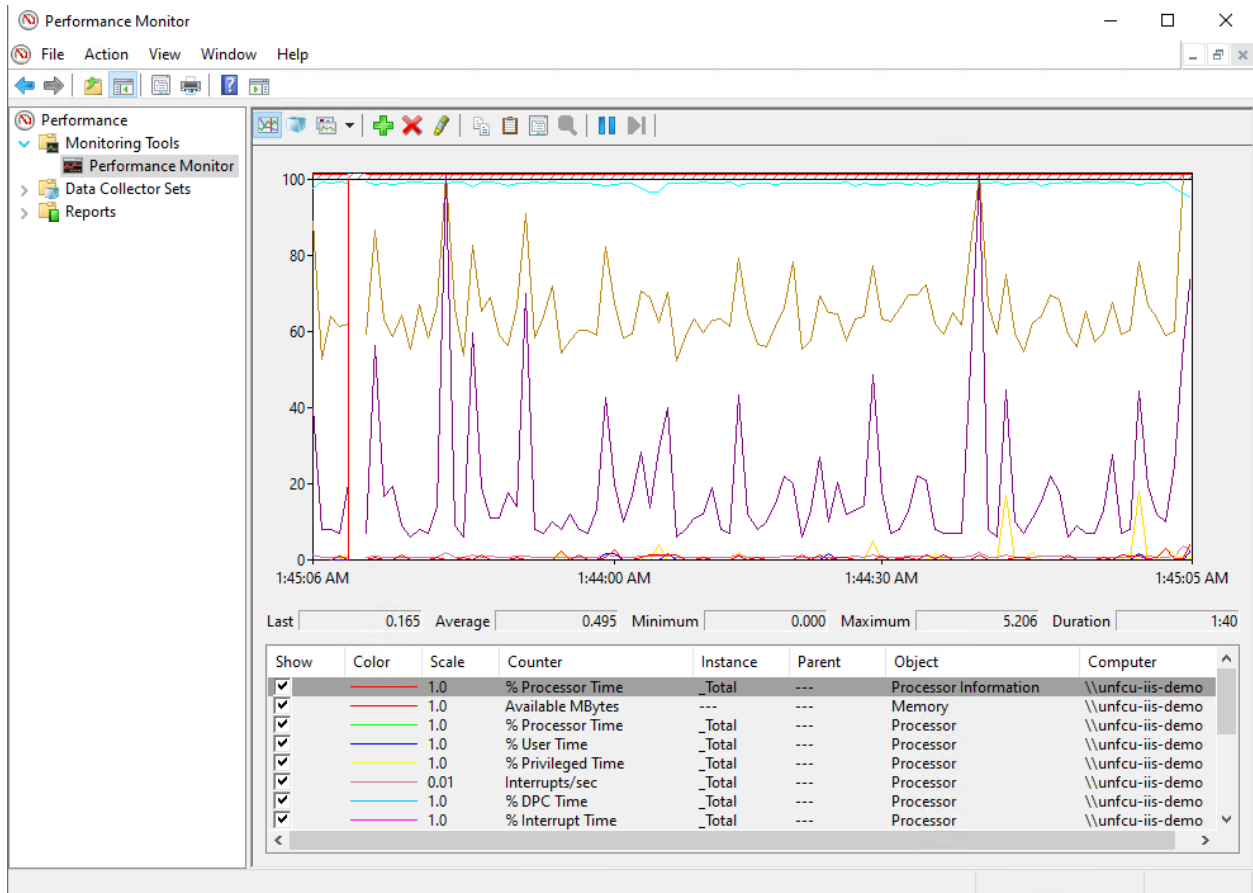
Failures: 3

Successes: 0

Avg Time: 5000ms

Performance During Incident

- CPU: 60% (baseline 5-10%)
- Memory: 5.5GB (from 7GB)
- Retry logic consuming resources



Recovery

Fix Applied

1. Updated web.config connection string
2. Changed hostname from .internal to .local
3. Restarted IIS Application Pool (DefaultAppPool)
4. Verified connectivity and authentication

Resolution Timeline

- ~1 minute: Detection (alert)
- ~2 minutes: Diagnosis (log analysis)
- ~2 minutes: Fix (config + restart)
- Total downtime: ~5 minutes

Successful Connection

Connection Status

Connection SUCCESSFUL

Database connected. Loan origination system is online and accepting requests.

Diagnostic Tests

 Test Connection (Broken)

 Apply Fix & Test

 Reset Test

Scenario 1: Click "Test Connection (Broken)" to simulate a database connection failure. This will generate multiple failed attempts in the IIS logs.

Scenario 2: After analyzing the logs and identifying the root cause, click "Apply Fix & Test" to correct the connection string and show successful recovery.

Connection Log Output

```
[2026-06-15 02:07:30] ===== TEST COMPLETED - CONNECTION FAILED =====
[2026-06-15 02:07:57] ===== RETESTING CONNECTION AFTER FIX =====
[2026-06-15 02:07:58] Connecting to: db-lending-prod-01.unfcu.local:5432
[2026-06-15 02:07:58] Attempting database connection...
[2026-06-15 02:07:59]  TCP connection established successfully
[2026-06-15 02:07:59] Authenticating with credentials...
[2026-06-15 02:08:00]  Authentication successful
[2026-06-15 02:08:00] Connected to database: lending_origination_db
[2026-06-15 02:08:00] Connection pool initialized: 5/20 connections
[2026-06-15 02:08:00] ===== CONNECTION TEST PASSED =====
[2026-06-15 02:08:00] System returned to normal operations
```

Detailed Error Analysis

✓ **Connection Status:** RESOLVED
Root Cause: Incorrect database hostname in configuration
Solution Applied: Updated hostname from .internal to .local
Resolution Time: 5 minutes

✓ **Resolution Summary:**

The database hostname was incorrectly configured as "db-lending-prod-01.unfcu.internal" when it should have been "db-lending-prod-01.unfcu.local". This DNS misconfiguration prevented all connection attempts from reaching the database server.

Fix Applied:

1. Updated connection string in web.config
2. Restarted IIS Application Pool "DefaultAppPool"
3. Verified connectivity to correct host
4. Confirmed database authentication successful

Recovery Verification:

- ✓ Database connection: SUCCESS
- ✓ Authentication: SUCCESS
- ✓ Connection pooling: ACTIVE (5/20 connections)
- ✓ Loan processing: ONLINE
- ✓ Credit check: ONLINE
- ✓ Member services: RESTORED

Status: FULLY OPERATIONAL

Incident Metrics:

Time to Detect: ~1 minute (automated monitoring alert)
Time to Diagnose: ~2 minutes (log analysis)
Time to Fix: ~2 minutes (configuration update + restart)
Total Downtime: ~5 minutes
Member Impact: 15 loan applications delayed

Performance Metrics

Attempts: 1

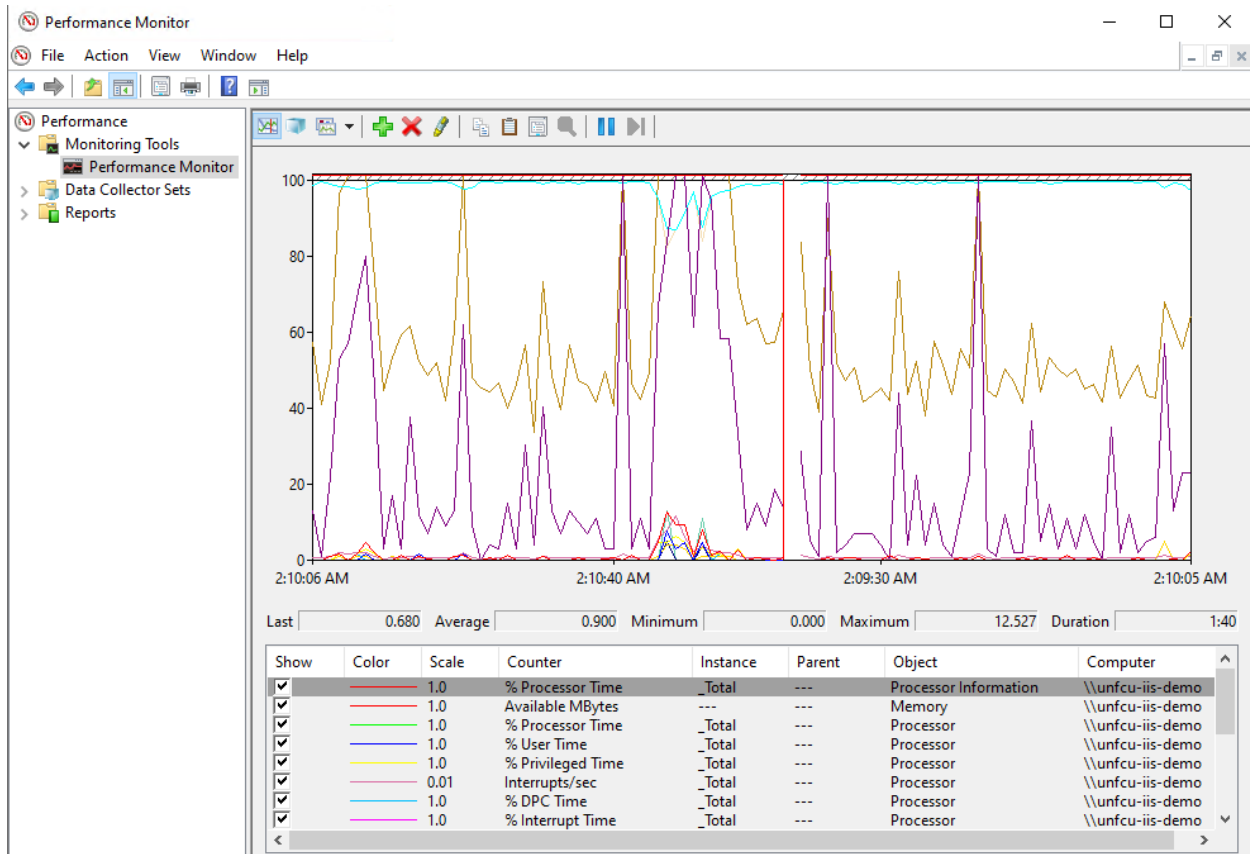
Failures: 0

Successes: 1

Avg Time: 400ms

Post-Recovery Performance

- CPU: 5-10% baseline
- Memory: 6-7GB restored
- Connection pool: 5/20 active
- Response time: 1-5ms
- Error rate: 0%



Log(Zoomed in) showcasing successful connection to the requested API.

Key Findings

Root Cause

Infrastructure domain migration (.internal → .local) not communicated to all teams. Application configuration retained outdated hostname, preventing DNS resolution.

Impact

- 15 member loan applications delayed
- Loan origination system unavailable
- Credit check service offline
- Mortgage processing blocked

Prevention

- Configuration validation in deployment pipeline
- DNS resolution tests pre-deployment
- Infrastructure change notifications to all teams
- Automated health checks on startup